

CS & IT ENGINEERING



Operating System

Process Synchronization

Lecture – 3

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Recap of Previous Lecture



doubt?

Topic

Critical Section

Topic

Solution of Critical Section Problem



Topics to be Covered



Topic

2-Process Solutions





Topic : Solution of Critical Section Problem

Mutual Exclusion:

If one process is executing the critical section, then other process is not allowed to enter into critical section.





Topic : Solution of Critical Section Problem

Progress:

If no any process is in critical section and any process wants to enter into critical section, then the process must be allowed.





Topic : Solution of Critical Section Problem

Bounded Waiting:

If a process p_1 is executing in critical section and other process p_2 is waiting for critical section, then the waiting time of p_2 must be bounded. Which means p_1 must not enter in to critical section again and again by keeping p_2 in waiting for long.



Topic : 2-Process Solution



entry section

C.S.

exit section

while (condition);



```
while (condition)
{
}
}
```

Solution 1

represents if C.S.
in locked or open

Boolean lock=false;

P0

```
while(true)
{
    while(lock);
    lock=true;
    //CS
    lock=false;
    RS;
}
```

P1

```
while(true)
{
    while(lock);
    lock=true;
    //CS
    lock=false;
    RS;
}
```

X Mutual Exclusion

✓ Progress

X Bounded waiting

(strict alternation)

Solution 2

✓ mutual Exclusion

✗ Progress

✓ Bounded waiting

int turn=0; 1
P0

```
while(true)
{
    while(turn!=0);
    CS
    turn=1;
    RS;
}
```

P0 can enter into CS when turn=0

P1

```
while(true)
{
    while(turn!=1);
    CS
    turn=0;
    RS;
}
```

P1 can enter into C.S. when turn=1

Peterson's Solution

to announce the willingness of process to enter into CS

0 1

Boolean Flag[2]={False, False};

int turn; ← for giving the priority

P0

```
while(true) {
    Flag[0]=true;
    turn=1;
    while(Flag[1] && turn==1);
        CS
    Flag[0]=False;
    RS;
}
```

P1

```
while(true){
    Flag[1]=true;
    turn=0;
    while(Flag[0] && turn==0);
        CS
    Flag[1]=False;
    RS;
}
```



2 mins Summary

Topic

Critical Section

Topic

Solution of Critical Section Problem





Happy Learning

THANK - YOU

